

Hospital Chatbot for Color Coded Clinical Pathways

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Outline of Presentation

- 1 Research gaps
- 2 Community Hospital case study
- 3 Transshipment optimization
- 4 NLP triage design
- 5 Conclusion

ED overcrowding and waiting for care



Source: Longdon (2026), Citi Newsroom.

<https://www.citinewsroom.com/2026/01/>

[queue-for-care-the-doctor-gap-bervelyn-longdon-writes/](https://www.citinewsroom.com/2026/01/queue-for-care-the-doctor-gap-bervelyn-longdon-writes/)



Source: Aminu (2021), Ghana Fact.

<https://ghanafact.com/>

[fact-check-korle-bu-teaching-hospital-overwhelmed-with-patients/](https://ghanafact.com/fact-check-korle-bu-teaching-hospital-overwhelmed-with-patients/)

Research gaps in prior work

We aim to infer acuity from free-form chief complaints to reduce ED overcrowding and waiting times (Mitchell et al., 2023; Yilmaz, 2021).

- ➊ **Vitals-first bias.** SATS and TEWS set acuity from bedside physiology, not from chief-complaint text when vitals are still unknown (South African Triage Group, 2012; Rominski et al., 2014; Stewart et al., 2023).
- ➋ **Pathway discontinuity.** Triage colours are assigned at the desk, but waiting-time and destination protocols are not consistently carried forward in digital workflows (Nurchis et al., 2025; Sundström et al., 2014).
- ➌ **Overcrowding without text pre-triage.** Priority-3/Green patients congest acute ED zones and extend waits for critical cases (Yilmaz, 2021; Mitchell et al., 2023).
- ➍ **Generative vs constrained NLP.** Open-ended conversational triage raises hallucination and audit concerns compared with fixed acuity-class outputs (Stewart et al., 2023; Cheng et al., 2025).
- ➎ **Class imbalance.** NLP triage papers often emphasize overall accuracy while under-reporting recall on rare high-acuity arrivals (Stewart et al., 2023; Berkowitz et al., 2019).

Case study setting

Community Hospital is a fictional Ghanaian LMIC facility (Kumasi-region style) (Rominski et al., 2014; Gyedu et al., 2016).

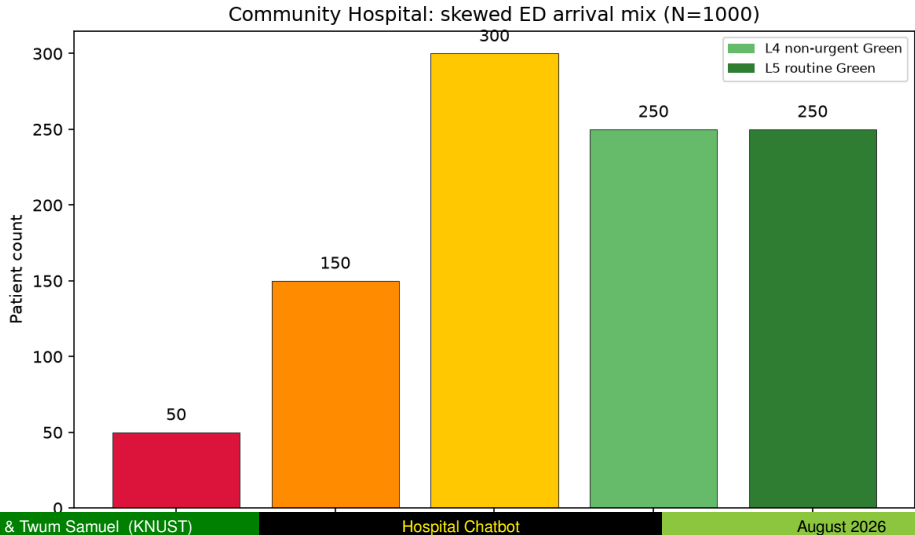
- Synthetic cohort: **N = 1000** patients with **skewed ED arrival mix** (5% / 15% / 30% / 25% / 25%) (Mitchell et al., 2023; This study, 2026).
- Ghanaian patient-voice chief complaints (Kejetia, Bantama, Ejisu, Sunyani, KATH).
- Ground-truth nurse labels $c \in \{1, \dots, 5\} \rightarrow C = f_{\text{SATS}}(c) \rightarrow \text{pathway } P(C)$ (South African Triage Group, 2012; Rominski et al., 2014).
- ~7% borderline complaints flagged for adjacent-level ambiguity (This study, 2026).
- Chatbot eval: **97.5%** colour-routing accuracy (975/1000); 8k holdout reference 99.92% (This study, 2026; Lee et al., 2020).

Dataset schema

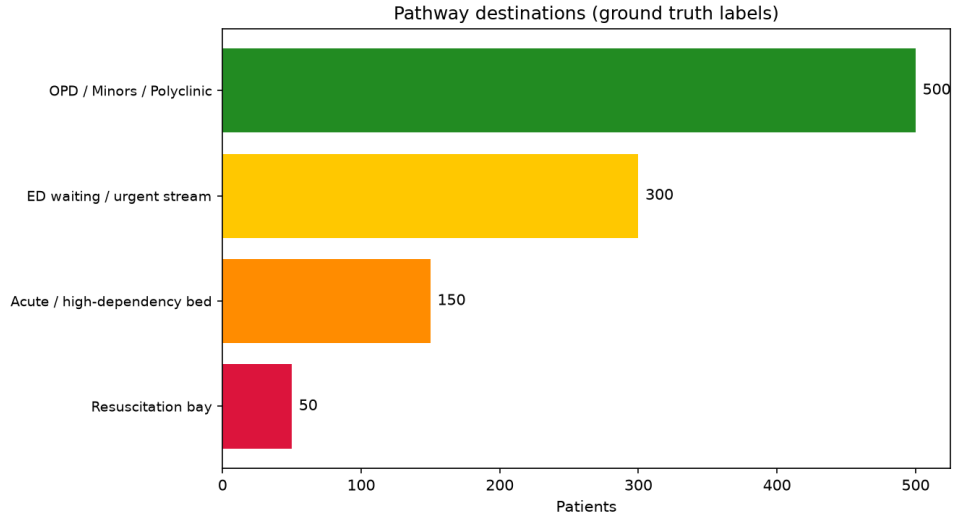
Field	Example
patient_id	CH-0042
sex / age	Female, 34
chief_complaint	“Crushing chest pain since Kejetia. . .”
acuity_level	2
sats_colour	Orange
pathway_destination	Acute / high-dependency bed
t_max_minutes	10
ambiguity_flag	true / false

Data: `community-hospital/data/community_hospital_patients.csv` (This study, 2026)

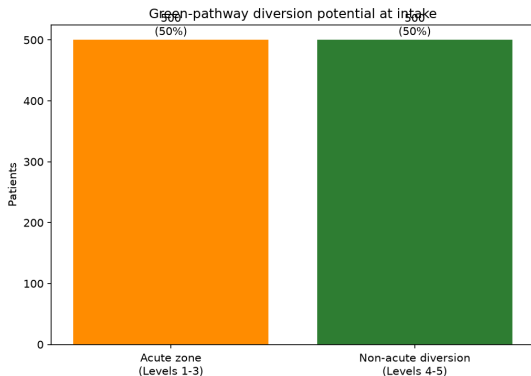
Acuity distribution



Pathway destinations



Green-pathway diversion

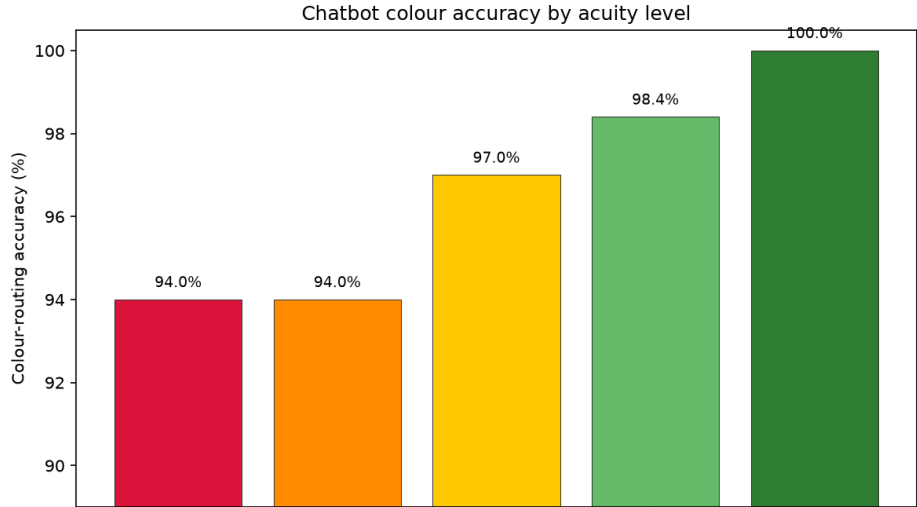


50% of cohort (levels 4–5, $n = 500$) routed to OPD / Minors / Polyclinic (This study, 2026).

Without early diversion, many Green patients queue in acute zones and worsen overcrowding (Yilmaz, 2021; Mitchell et al., 2023).

Chatbot assigns Green at intake, before the physical desk (Sundström et al., 2014; This study, 2026).

Chatbot colour-routing accuracy



Case study conclusion

Findings

- NLP maps Ghanaian free text to SATS colour before vitals are available (Stewart et al., 2023; Gligorijević et al., 2018; This study, 2026).
- Every case includes pathway destination and $T_{\max}$ (Sundström et al., 2014; This study, 2026).
- 500/1000 patients are Green diversion candidates: operational lever on crowding (Yilmaz, 2021; This study, 2026).

Simple meaning

Next: transshipment LP model for ED patient flow (This study, 2026).

ED overcrowding as a network problem

Problem: long waits and acute-zone congestion when non-urgent patients enter the wrong queue (Yilmaz, 2021; Mitchell et al., 2023).

OR tool: transshipment LP

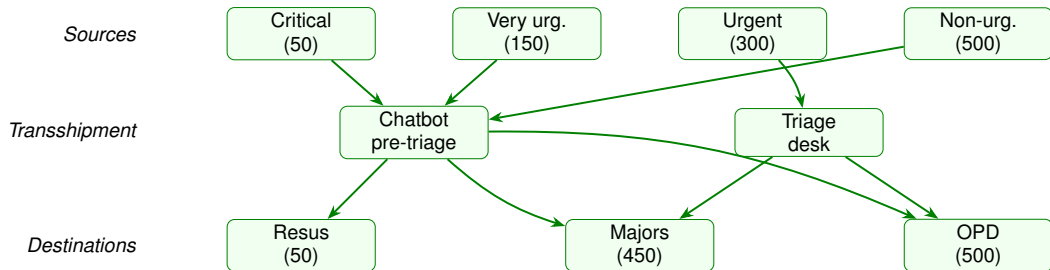
$$\min Z = \sum_{(i,j)} c_{ij} x_{ij}$$

subject to supply, balance, and demand constraints.

Simple meaning

Patients are flow; arcs are routing paths; cost c_{ij} is expected wait plus wrong-queue penalty (This study, 2026).

Hospital transshipment network



Sources (1–4) → transshipment nodes (chatbot, desk) → destinations (This study, 2026; Yilmaz, 2021).

LP structure

Decision variables: x_{ij} = patients routed on arc (i, j) .

Supply: arrivals (50, 150, 300, 500) (This study, 2026).

Balance: inflow = outflow at chatbot and desk.

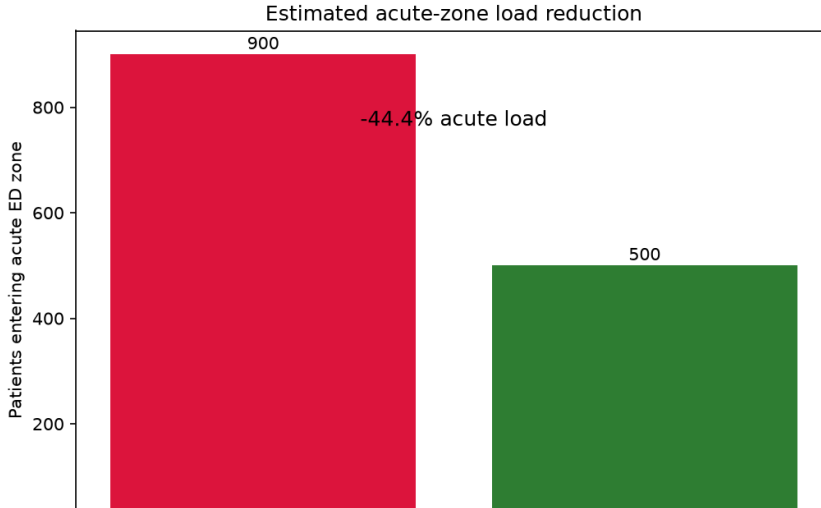
Demand: resuscitation 50, majors 450, OPD 500 (This study, 2026).

Community Hospital instance

Scenario	Total cost Z	Acute-zone load
Without chatbot	7150	900
With chatbot	2700	500
Reduction	62%	44%

Results from synthetic cohort routing (This study, 2026).

Acute-zone load reduction



Conversational NLP supports triage

- Chat interface collects chief complaint in the patient's own words (Ghanaian English, local context) (Stewart et al., 2023; An et al., 2025).
- Optional: voice intake (Twi $\rightarrow$ English), medical gate, OpenMed entity tags (Lee et al., 2020; This study, 2026).
- Conversation clarifies symptoms, but **output is not free-form medical advice** (Cheng et al., 2025; Stewart et al., 2023).

Fixed outputs

- 1 Predicted acuity level $\hat{c} \in \{1, \dots, 5\}$ (Gligorijević et al., 2018; Lee et al., 2020)
- 2 SATS colour $C = f_{\text{SATS}}(\hat{c})$ (South African Triage Group, 2012; Rominski et al., 2014)
- 3 Pathway card $P(C)$: destination, T_{max} , actions, escalation (Sundström et al., 2014)

Versus generative clinical chatbots

Generative LLM	Curatio (this project)
Open-ended text reply (An et al., 2025)	Fixed class + pathway card (Stewart et al., 2023)
Hallucination risk (Stewart et al., 2023)	Bounded output set (This study, 2026)
Hard to audit (Cheng et al., 2025)	Confidence + per-level metrics (This study, 2026)
“You might have. . .”	“Orange → acute bed, 10 min” (South African Triage Group, 2012)

Simple meaning

Nurse remains in the loop. Chatbot pre-triages and routes; it does not replace clinical judgment (Cheng et al., 2025; Rominski et al., 2014).

Summary

- 1 **Research gaps:** estimate $\hat{c}$ from X when $\mathbf{v} = \emptyset$; bind C to $P(C)$; model ED flow as transshipment LP (Stewart et al., 2023; Sundström et al., 2014; Mitchell et al., 2023).
- 2 **Community Hospital:** 1000 skewed cases; 97.5% colour accuracy; pathway cards on every row (This study, 2026; South African Triage Group, 2012).
- 3 **Transshipment OR:** 62% cost reduction, 44% acute-load reduction with chatbot Green diversion (This study, 2026; Yilmaz, 2021).
- 4 **NLP design:** conversational intake, constrained output (not free-form diagnosis) (Stewart et al., 2023; Lee et al., 2020; Cheng et al., 2025).

Limitations

- Synthetic cohort: not yet prospective KATH validation (This study, 2026; Rominski et al., 2014).
- Simulated chatbot eval when ML service offline (8k holdout 99.92% reference) (This study, 2026; Lee et al., 2020).
- Transshipment is an analytical model, not a deployed optimizer (This study, 2026; Mitchell et al., 2023).

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